From COVID-19 Vaccine Hesitancy to Mpox: A Comparative Study of Stance Labelling and Machine-Learning Classification

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Scientific synthesis of published collaborative research

Abstract. Social-media discourse provides a valuable but linguistically complex source for examining public attitudes during emerging health crises. This report synthesises two published studies that investigated sentiment and stance classification across COVID-19 vaccine hesitancy and the 2022 Mpox outbreak, with equal emphasis on human labelling and cross-crisis model transfer. The COVID-19 study constructed a manually labelled corpus of 30,000 South African vaccine-related tweets and compared support vector machine, LSTM, Bi-LSTM, BERT and RoBERTa classifiers, with RoBERTa achieving the strongest three-class F1-score of 61%. Removing the ambiguous neutral class increased both BERT and RoBERTa performance to approximately 80% F1. The subsequent Mpox study retained the established human-labelling framework while evaluating VADER, TextBlob and four transformer models on 20,604 Mpox-related tweets. COVID-19-adapted BERT and RoBERTa models outperformed generic pretrained alternatives before Mpox-specific fine-tuning, reaching 45% F1 and 47% accuracy, before increasing to 68% and 69% F1, respectively, following fine-tuning on manually labelled Mpox data. Lexicon-based labelling showed substantially weaker and class-dependent performance. Together, the studies demonstrate that knowledge learned from related health discourse can transfer across crises, but that such transfer remains incomplete without new context-specific supervision. Reliable human labelling and domain-adapted transformer modelling therefore emerge as complementary components of robust stance and sentiment classification in rapidly evolving public-health discourse. [1, 2]

Keywords: COVID-19; Mpox; vaccine hesitancy; stance detection; sentiment analysis; manual labelling; natural language processing; machine learning; BERT; RoBERTa; transfer learning; social media

1 Introduction

Health emergencies are accompanied not only by epidemiological and clinical challenges, but also by rapidly evolving public attitudes, misinformation, uncertainty and disagreement surrounding public-health interventions. Social-media platforms provide a large and temporally responsive source of user-generated discourse through which these attitudes can be studied using natural language processing (NLP) and machine learning (ML). Within this setting, sentiment analysis characterises the emotional polarity expressed in text, whereas stance detection focuses more specifically on whether an author supports, opposes or remains neutral towards a defined target or proposition. Distinguishing these related tasks is particularly important in health-related discourse, where the polarity of individual words does not necessarily correspond to the position ultimately expressed by the author. [3, 13]

COVID-19 vaccine hesitancy provided an important setting in which to examine these challenges. Although social-media data were increasingly used to study public responses to vaccination during the pandemic, comparatively little computational work focused specifically on South African user-generated content, and much of the wider literature relied on automatically generated sentiment labels. This was problematic in a setting where local slang, informal language, sarcasm and context-dependent expressions could influence the interpretation of a post. The first study addressed this gap by constructing a corpus of 30,000 South African vaccine-related tweets that were manually labelled as positive, neutral or negative, and subsequently used to evaluate classical machine-learning, recurrent neural-network and transformer-based classifiers. The emphasis on manual labelling was therefore not incidental to the modelling pipeline, but formed a central methodological component for establishing context-sensitive reference labels against which model behaviour could be assessed. [1, 14, 15]

The subsequent Mpox study extended this methodological foundation to a different emerging public-health crisis. Rather than assuming that a labelling or classification strategy developed for COVID-19 would generalise unchanged, the study tested the extent to which earlier methods remained useful when applied to a new topic, a new time period and a differently distributed stance corpus. The Mpox analysis retained manual stance labelling as the reference standard, evaluated lexicon-based tools such as VADER and TextBlob against those labels, and assessed transformer models that had previously been adapted to COVID-19 discourse before further fine-tuning them on Mpox data. In this way, the second study provided a direct test of both methodological continuity and the limits of cross-crisis transferability. [2, 6, 7]

This report synthesises the two published studies within a single comparative framework, with equal emphasis on the construction of reliable human-labelled data and on the transferability of machine-learning methods between related but distinct public-health crises. Rather than treating the COVID-19 and Mpox analyses as independent case studies, their data collection, labelling procedures, modelling choices and evaluation strategies are examined concurrently to identify methodological continuity, necessary adaptation and differences in performance. The central questions are therefore whether context-sensitive human labelling provides a sufficiently reliable foundation for stance and sentiment classification, and to what extent models developed using COVID-19 vaccine-related discourse retain useful predictive information when transferred to Mpox before further domain-specific fine-tuning. The synthesis is restricted to methods and results reported in the original IEEE and PLOS Digital Health publications; no new model training or experimental results are introduced here. [1, 2]

The wider literature supports the use of social-media data as a complementary source of information for public-health research, while also emphasising important limitations related to representativeness, validation and the interpretation of rapidly changing online discourse. NLP and machine-learning methods can make large volumes of such text analytically tractable, but their usefulness depends on whether the categories being predicted correspond meaningfully to the attitudes expressed by users. This concern is especially relevant to stance detection, for which the relationship between textual sentiment and position towards a specific target is not necessarily direct. Previous reviews have consequently identified target definition, contextual information, domain dependence and reliable benchmark datasets as persistent challenges in social-media stance classification. These considerations motivate the comparative analysis undertaken here, in which human-labelled corpora provide the empirical foundation for examining both model performance and methodological transfer between COVID-19 vaccine hesitancy and Mpox discourse. [3, 13]

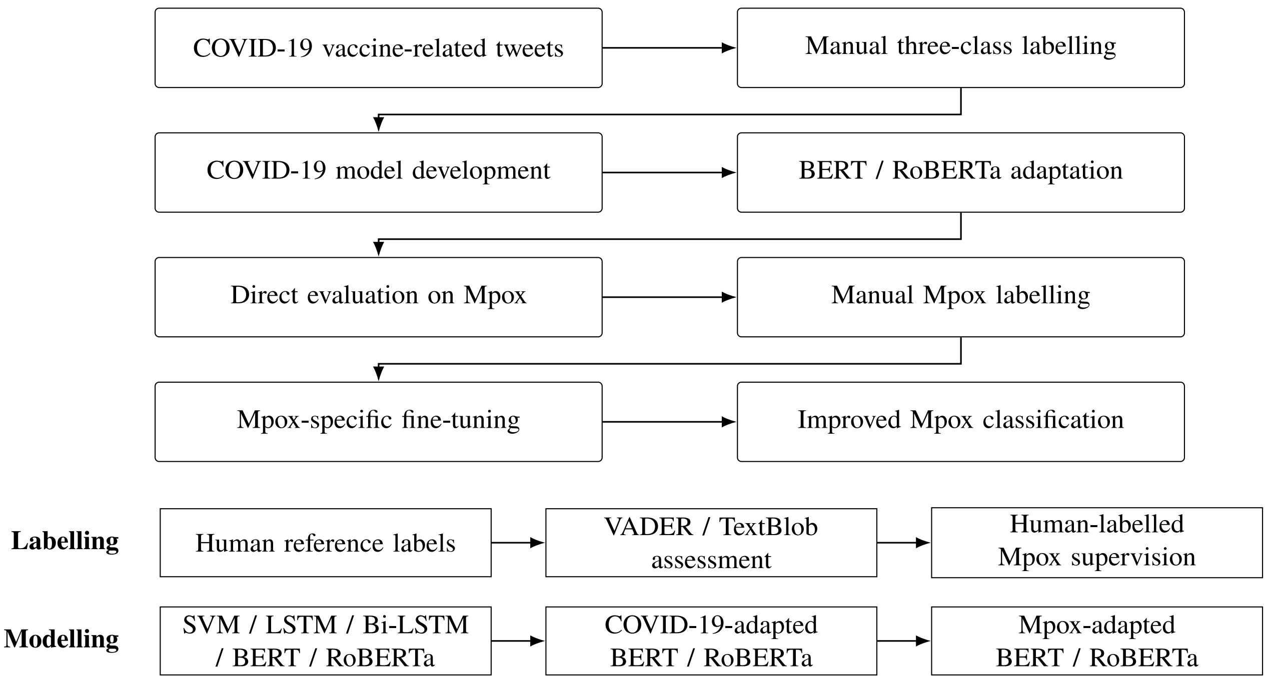

Figure 1: Methodological progression from COVID-19 vaccine-hesitancy analysis to Mpox stance classification, illustrating the continuity of human labelling and model adaptation across the two studies.

2 Methods

2.1 Data Collection

Both studies used Twitter data collected under the Twitter Research License and followed a comparable sampling strategy, providing a consistent foundation for comparison across the two public-health contexts. The COVID-19 study began with 100,000 geotagged tweets from South African users collected between 5 March 2020, when the first COVID-19 case was identified in South Africa, and 24 November 2021, when the Omicron variant was first detected in the country. Extraction was performed using vaccine- and vaccination-related keywords and hashtags, after which cluster sampling was used to select 30,000 unique tweets for manual labelling and subsequent analysis. The Mpox study similarly began with approximately 100,000 collected tweets and applied the cluster-sampling procedure established in the COVID-19 work to select 20,604 tweets for manual labelling. In this case, extraction focused on Mpox-related hashtags between 1 May and 5 September 2022, spanning the period around the peak in global infections during the 2022 outbreak. Thus, although the two corpora addressed different crises and populations of discourse, their construction retained an explicit methodological link through the use of Twitter-derived data and a common sampling framework. [1, 2]

The COVID-19 corpus was geographically constrained to South African users and was designed to capture vaccine-related discourse across the country's nine provinces. Data extraction was conducted in monthly intervals and used 15 vaccine-related keywords or hashtags per query, including terms associated with specific vaccine products, vaccination status and pro- or anti-vaccine attitudes. The available Twitter metadata included geolocation and other contextual information, allowing the sampling procedure to retain the explicitly South African focus of the study. The Mpox dataset differed in scope: it was constructed around outbreak-related hashtags rather than a geographically restricted South African population, reflecting the broader public discourse surrounding the 2022 outbreak. Because of Twitter's data-sharing policy, the published Mpox supplementary dataset contains only tweet identifiers and their manually assigned labels; the underlying tweet text and metadata must be recovered by hydrating those identifiers. These differences are important for later comparison because the two datasets share a broadly similar collection and sampling lineage but represent distinct geographic and topical populations. [1, 2]

Despite their shared sampling lineage, the two corpora should not be interpreted as directly matched datasets. The COVID-19 corpus was explicitly defined around South African users and vaccine-related discourse, whereas the Mpox corpus captured a broader set of posts centred on the 2022 outbreak and the public response to it. The collection periods, geographic scopes, search terms and underlying public-health contexts therefore differ substantially. These distinctions are methodologically important: they allow the later Mpox analysis to test whether information learned from COVID-19 discourse remains useful under a shift in topic and data distribution, but they also prevent simple numerical comparisons between the two datasets from being interpreted as controlled head-to-head experiments. The common sampling and labelling lineage consequently provides methodological continuity without implying equivalence between the underlying populations of social-media discourse.

Characteristic	**COVID-19 study**	**Mpox study**
Platform	Twitter	Twitter
Initial collection	Approximately 100,000 tweets	Approximately 100,000 tweets
Final labelled corpus	30,000	20,604
Collection dates	5 Mar 2020–24 Nov 2021	1 May–5 Sep 2022
Geographic scope	South Africa	Broader outbreak-related discourse
Topic	COVID-19 vaccination / vaccine hesitancy	2022 Mpox outbreak
Sampling	Cluster sampling	Cluster sampling based on the same framework
Labels	Manual, three-class	Manual, three-class
Positive	31.7%	22.2%
Neutral	36.1%	35.3%
Negative	32.2%	42.5%

Table 1: Comparison of the COVID-19 and Mpox social-media datasets used in the two published studies.

2.2 Labelling Methodology

Manual labelling formed the reference framework in both studies. In the COVID-19 work, a team of 11 annotators classified the 30,000 selected tweets as positive, negative or neutral according to the author's stance towards COVID-19 vaccination. The central question was whether the author approved or disapproved of receiving a vaccine, with annotators instructed to consider not only word choice but also punctuation, grammar, symbols, tone, sarcasm and local slang. The labelling scheme was based on an operationalisation of vaccine hesitancy in which overt hesitancy referred to explicit refusal or strong opposition to vaccination and was assigned a negative label, whereas subtle hesitancy referred to indecision or delayed willingness to vaccinate and was assigned a neutral label. Positive labels represented clear approval and willingness to vaccinate, while tweets whose stance was unclear or unrelated were also assigned to the neutral class by default. The Mpox study retained this human-labelled, three-class approach as its methodological foundation, allowing automated labelling and transferred transformer models to be evaluated against labels derived through contextual human interpretation rather than automatically assigned polarity. [1, 2, 4]

Intercoder reliability was explicitly quantified in the COVID-19 study using a multistage consensus procedure. The 30,000 tweets were divided into sets of 1,000 and assigned to randomly paired annotators, who labelled each set independently before Cohen's kappa (κ) was calculated to measure agreement beyond chance. Disagreements were reviewed with a third team member, additional rules were introduced where recurring ambiguities emerged, and the process was repeated over six iterations for each group. The mean κ score increased from 0.67 initially to 0.85 after iterative discussion and rule refinement, indicating a substantial improvement in label consistency. The Mpox study then reused the same hand-labelling rules and procedures established in the COVID-19 work, preserving this methodological framework while redefining the stance target around trust, concern and perceived government control of the 2022 Mpox outbreak. [1, 2, 5]

Both studies also used the same qualitative framework to examine where automated labelling diverged from human interpretation. VADER and TextBlob predictions were compared against the manually assigned labels, with examples organised into four categories: clear-cut cases, for which the intended stance or sentiment was unambiguous; borderline cases, where either of two adjacent labels could reasonably be assigned; difficult-to-label cases, containing simultaneously strong positive and negative cues; and same-text cases, in which punctuation or emoji differences altered the intended meaning despite otherwise identical wording. This framework was first applied in the COVID-19 study to demonstrate the limitations of automated polarity-based classification and was subsequently retained in the Mpox study, allowing the same sources of linguistic ambiguity to be examined across both public-health contexts. [1, 2, 6, 7]

Element	**COVID-19 framework**	**Mpox framework**
Reference labels	Manual	Manual
Classes	Positive / Neutral / Negative	Positive / Neutral / Negative
Primary target	Vaccine sentiment / hesitancy	Stance towards Mpox outbreak / response
Positive	Clear approval / willingness to vaccinate	Supportive stance
Neutral	Subtle hesitancy, indecision or unclear stance	Neutral / uncertain stance
Negative	Overt hesitancy, refusal or opposition	Negative / oppositional stance
Context considered	Slang, sarcasm, punctuation, emojis and tone	Same contextual principles retained
Reliability procedure	Iterative paired annotation and consensus	Framework reused from COVID-19 study
Cohen's κ	0.67 initially; 0.85 after refinement	Not separately re-reported
Automated comparison	VADER and TextBlob	VADER and TextBlob
Error categories	Clear-cut; borderline; difficult; same-text	Same four categories

Table 2: Comparison of the human-labelling frameworks and automated labelling assessments used for COVID-19 vaccine hesitancy and Mpox discourse.

2.3 Modelling Approach

The COVID-19 study established the principal modelling framework subsequently carried into the Mpox analysis. Five supervised classification approaches were evaluated on the manually labelled COVID-19 corpus: support vector machine (SVM), long short-term memory (LSTM), bidirectional LSTM (Bi-LSTM), BERT-base-cased and RoBERTa-base. Two alternative preprocessing strategies were also compared. The corpus-based approach applied conventional text normalisation, whereas the semantics-based approach retained more contextual information that could influence interpretation. Hyperparameters for the candidate models were tuned using the Weights & Biases platform before final evaluation. This combination of classical machine learning, recurrent neural networks and transformer architectures allowed the study to compare models with progressively greater capacity for representing contextual information in informal social-media text. [1, 8–11]

The Mpox study narrowed this modelling framework to transformer-based approaches in order to examine cross-crisis transfer more directly. Two models originating from the earlier COVID-19 work–COVID-19 BERT and

COVID-19 RoBERTa–were evaluated alongside two independently pretrained sentiment models, NLP-Town BERT and Cardiff-NLP RoBERTa. All four models were first tested on the held-out Mpox data without Mpox-specific fine-tuning, after which they were fine-tuned on the manually labelled Mpox training set and evaluated again on the same test set. The COVID-19 BERT and RoBERTa models retained the hyperparameter values established in the earlier study, while the two external pretrained models used their own specified learning rates, batch sizes and epoch counts. Unlike the COVID-19 analysis, no additional preprocessing was applied before transformer training because each model used its own built-in tokenisation procedure. This design allowed the Mpox study to distinguish performance gained from prior adaptation to related health discourse from the additional benefit obtained through direct fine-tuning on newly labelled Mpox data. [2, 10, 11]

Model optimisation was treated as a substantive part of the COVID-19 modelling workflow rather than relying on default configurations. All candidate models underwent extensive hyperparameter tuning using Bayesian optimisation through the Weights & Biases (WandB) platform, with model-specific search spaces reflecting their different architectures. For SVM, the search varied the kernel, γ and regularisation cost C; the LSTM and Bi-LSTM searches varied architectural and training parameters including dropout, weight decay, learning rate, batch size, dense, embedding and hidden dimensions, number of epochs and optimiser choice. The transformer search was substantially narrower because BERT and RoBERTa were already pretrained and required downstream fine-tuning rather than training from scratch; the published search space included learning rates from 10^{-6} to 10^{-4}, 2–12 epochs and batch sizes from 8–64, with AdamW retained as the standard optimiser. The broader recurrent-model search spaces reflected their greater number of trainable architectural choices, whereas the transformer optimisation concentrated on the parameters governing fine-tuning. The resulting COVID-19 BERT and RoBERTa configurations were subsequently carried forward into the Mpox study, creating a direct methodological link between model optimisation in the first study and transfer evaluation in the second. [1, 2, 16]

Model / stage	**Preprocessing**	**Selected configuration reported in the source study**
SVM	Semantics and lexical	RBF kernel; $\gamma = 0.1$; $C = 4$
LSTM	Semantics	dropout = 0.5863; learning rate = 0.8495; batch = 399; dense = 804; embedding = 393; hidden = 462; epochs = 96; weight decay = 0.8944; optimiser = Adam
Bi-LSTM	Semantics	dropout = 0.3354; learning rate = 0.1515; batch = 623; dense = 623; embedding = 852; hidden = 194; epochs = 191; weight decay = 0.8655; optimiser = RMSprop
LSTM	Lexical	dropout = 0.1636; learning rate = 0.5612; batch = 262; dense = 776; embedding = 692; hidden = 352; epochs = 66; weight decay = 0.7516; optimiser = Adam
Bi-LSTM	Lexical	dropout = 0.3267; learning rate = 0.5735; batch = 447; dense = 447; embedding = 831; hidden = 82; epochs = 293; weight decay = 0.7729; optimiser = Adam
BERT-base-cased	COVID-19 fine-tuning	dropout = 0.1; learning rate = 8.841×10^{-5}; batch = 32; epochs = 4; AdamW; weight decay not tuned
RoBERTa-base	COVID-19 fine-tuning	dropout = 0.1; learning rate = 5.576×10^{-5}; batch = 36; epochs = 7; AdamW; weight decay not tuned
COVID-19 BERT	Mpox fine-tuning	inherited COVID-19 learning rate = 8.841×10^{-5}; batch = 32; epochs = 4; weight decay = 0
COVID-19 RoBERTa	Mpox fine-tuning	inherited COVID-19 learning rate = 5.576×10^{-5}; batch = 36; epochs = 7; weight decay = 0
NLP-Town BERT	Mpox fine-tuning	learning rate = 1.0×10^{-4}; batch = 32; epochs = 8; weight decay = 0
Cardiff-NLP RoBERTa	Mpox fine-tuning	learning rate = 1.0×10^{-3}; batch = 64; epochs = 10; weight decay = 0

Table 3: Selected optimisation configurations reported for the COVID-19 models and retained or specified for Mpox fine-tuning. Values for the COVID-19 models are taken from the peer-reviewed IEEE configuration table; the Mpox study explicitly reused the COVID-19 transformer settings and specified the two external-model settings.

2.4 Evaluation Framework

Both studies evaluated classification performance using precision, recall, F1-score and accuracy, with results reported both overall and, where applicable, for the positive, neutral and negative classes separately. In the COVID-19 study, F1-score provided the principal basis for comparing the SVM, LSTM, Bi-LSTM, BERT and RoBERTa classifiers, while the transformer models were additionally compared with their corresponding pretrained configurations and with VADER and TextBlob. The later IEEE version also applied non-parametric Wilcoxon signed-rank tests to the class-level F1-scores to assess whether selected performance differences were statistically significant. In the Mpox study, the manually assigned labels again served as the reference: VADER and TextBlob were evaluated directly against these labels, while the four transformer models were tested on the held-out Mpox set both before and after Mpox-specific fine-tuning. The Mpox paper reports that one-fifth of the corpus was allocated for testing and the remainder for training, thereby allowing the effect of domain-specific fine-tuning to be assessed against a common evaluation set. [1, 2]

Both studies supplemented aggregate performance metrics with targeted diagnostic analyses designed to explain model failure rather than report classification scores alone. In the COVID-19 study, BERT and RoBERTa were re-evaluated after removing the neutral class, allowing the effect of three-way classification difficulty to be isolated; the resulting two-class models improved to approximately 80% F1, compared with 60–61% in the original three-class setting. In the Mpox study, Latent Dirichlet Allocation (LDA) was applied to tweets misclassified by the COVID-19 RoBERTa model before and after Mpox-specific fine-tuning. Five coherent topics were identified in each case, and the comparison was used to determine which error themes persisted, disappeared or became more specific after adaptation to the new domain. Together, these analyses extended evaluation beyond headline accuracy and F1 values by examining the linguistic and topical structure underlying classification errors. [1, 2, 12]

3 Results

3.1 Label Distributions

The manually labelled datasets differed noticeably in class composition. The COVID-19 corpus was comparatively balanced, with 31.7% of tweets labelled positive, 36.1% neutral and 32.2% negative; this distribution was sufficiently even that no additional class-balancing procedure was required before model training. By contrast, the Mpox corpus contained 22.2% positive, 35.3% neutral and 42.5% negative tweets, producing a clearer shift towards the negative class while retaining a similar proportion of neutral examples. The difference in class composition is relevant to the later model comparison because it indicates that the Mpox task was not simply a repetition of the COVID-19 classification problem on a new topic; the relative prevalence of the three stance classes also changed between the two crises. [1, 2]

3.2 Model Performance

The COVID-19 results showed a clear performance hierarchy across the five evaluated model families. Among the non-transformer approaches, LSTM achieved an overall F1-score of 49%, Bi-LSTM 52%, and SVM 54%, making SVM the strongest of the conventional and recurrent models. Fine-tuned BERT-base-cased and RoBERTa-base performed better, reaching overall F1-scores of 60% and 61%, respectively, with RoBERTa therefore providing the best three-class performance in the study. The improvement of the transformer models is consistent with their greater ability to represent contextual information in short, informal and linguistically complex social-media text, although the absolute scores also show that the task remained difficult despite extensive model optimisation. [1, 10, 11]

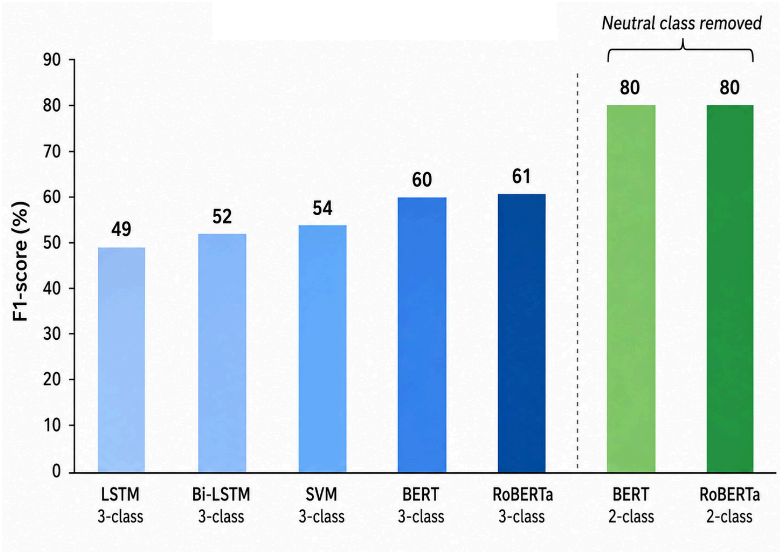

Figure 2: Overall F1-score of classifiers evaluated on the three-class COVID-19 vaccine-hesitancy task. BERT and RoBERTa achieved the strongest performance; removing the neutral class increased both transformer models to approximately 80% F1 in the corresponding two-class experiment.

The Mpox results provided direct evidence of useful, but incomplete, cross-crisis transfer. Before Mpox-specific fine-tuning, the generic NLP-Town BERT and Cardiff-NLP RoBERTa models achieved overall F1-scores of 40% and 39%, with accuracies of 39% and 40%, respectively. By comparison, the COVID-19 BERT and COVID-19 RoBERTa models both reached an F1-score of 45% and an accuracy of 47% when applied directly to Mpox, indicating that prior adaptation to COVID-19 health discourse retained information useful for the new task. The largest gains occurred after fine-tuning on the manually labelled Mpox corpus: COVID-19 BERT increased to 68% F1 and accuracy, while COVID-19 RoBERTa achieved the strongest overall performance at 69% F1 and 69% accuracy. The progression from generic pretrained models, through COVID-19-adapted models, to Mpox-specific fine-tuning therefore shows that related prior-domain training provided a transferable advantage, while substantial additional improvement still depended on labelled data from the new health context. [2]

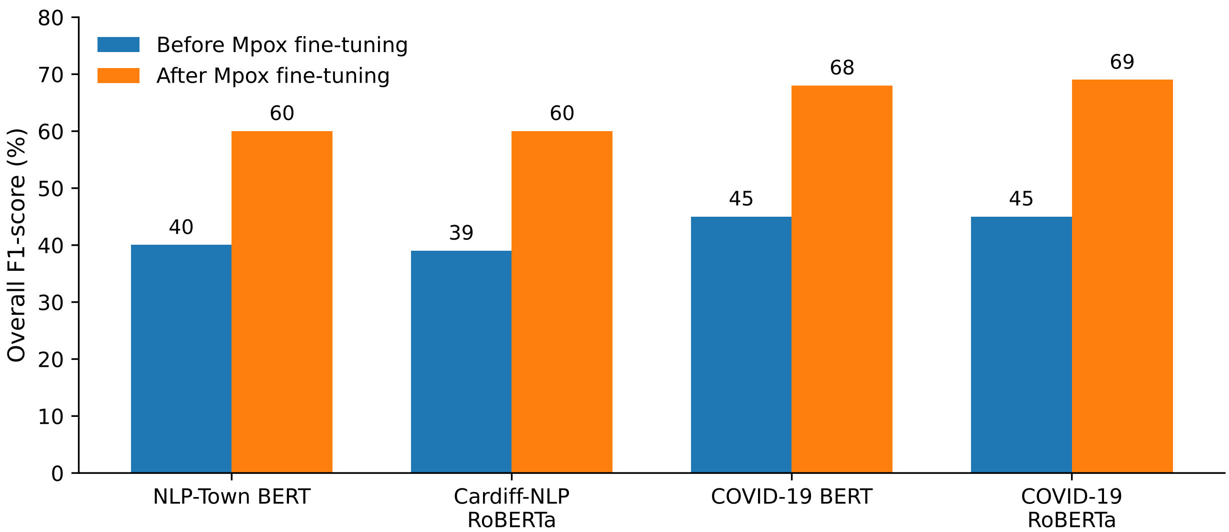

Figure 3: Transformer performance on the Mpox stance-classification task before and after Mpox-specific fine-tuning. Models previously adapted to COVID-19 discourse provided a stronger starting point than generic pretrained alternatives, while fine-tuning on manually labelled Mpox data produced the highest overall F1-scores.

The Mpox results also quantified the limitations of lexicon-based automated labelling when evaluated against the manually assigned stance labels. VADER achieved an overall precision of 49%, recall of 50%, F1-score of 47% and accuracy of 50%, while TextBlob achieved corresponding values of 41%, 40%, 37% and 40%. Performance was uneven across classes: VADER attained 71% recall for neutral tweets but only 20% for positive tweets, whereas TextBlob reached 62% recall for the neutral class but only 16% for positive tweets. These results indicate that the automated methods did not simply make uniformly distributed errors, but exhibited class-dependent behaviour that could distort the resulting stance distribution. Although VADER initially exceeded the accuracy of the four transformer models before Mpox-specific fine-tuning, every transformer model surpassed it after training on the manually labelled Mpox corpus, with the best-performing COVID-19 RoBERTa model reaching 69% accuracy and F1-score. [2, 6, 7]

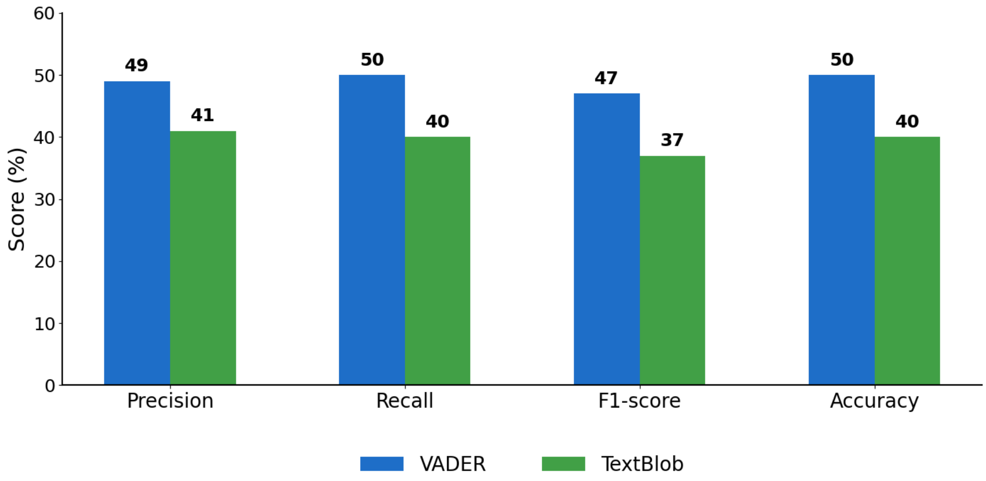

Figure 4: Performance of VADER and TextBlob against manually assigned Mpox stance labels. Both lexicon-based approaches showed limited overall performance and substantial class-dependent variation, particularly for positive examples.

3.3 Comparative Performance

Across the two studies, performance depended strongly on both the definition of the classification problem and the availability of domain-specific labelled data. In the COVID-19 analysis, removing the neutral class increased BERT and RoBERTa performance from approximately 60–61% F1 in the three-class task to about 80% F1 in the resulting positive-versus-negative classification, demonstrating that neutral and subtly hesitant examples accounted for a substantial proportion of the classification difficulty. In the Mpox analysis, a different effect was observed: retaining the same three-class formulation but introducing Mpox-specific labelled examples increased the COVID-19 BERT and RoBERTa models from 45% F1 before fine-tuning to 68% and 69%, respectively, after fine-tuning. These results are not directly comparable as a single benchmark because the datasets and evaluation settings differ, but together they show that model performance was sensitive both to class ambiguity and to adaptation to the linguistic context of the target health crisis. [1, 2]

The COVID-19 study further showed that the observed gains from model adaptation were statistically meaningful rather than simple fluctuations in class-level performance. Wilcoxon signed-rank testing found significant differences between fine-tuned RoBERTa and pretrained RoBERTa ($p < 0.001$), between fine-tuned BERT and pretrained BERT ($p < 0.001$), and between SVM and fine-tuned RoBERTa ($p < 0.01$). The Mpox analysis complemented this quantitative comparison with topic modelling of RoBERTa errors before and after Mpox-specific fine-tuning. LDA identified five coherent topics in each misclassification set, allowing the authors to examine which error themes persisted, disappeared or became more narrowly defined after adaptation. Together, these results indicate that fine-tuning affected not only aggregate performance metrics but also the structure of the remaining classification errors, reinforcing the importance of domain-specific adaptation across both studies. [1, 2, 12]

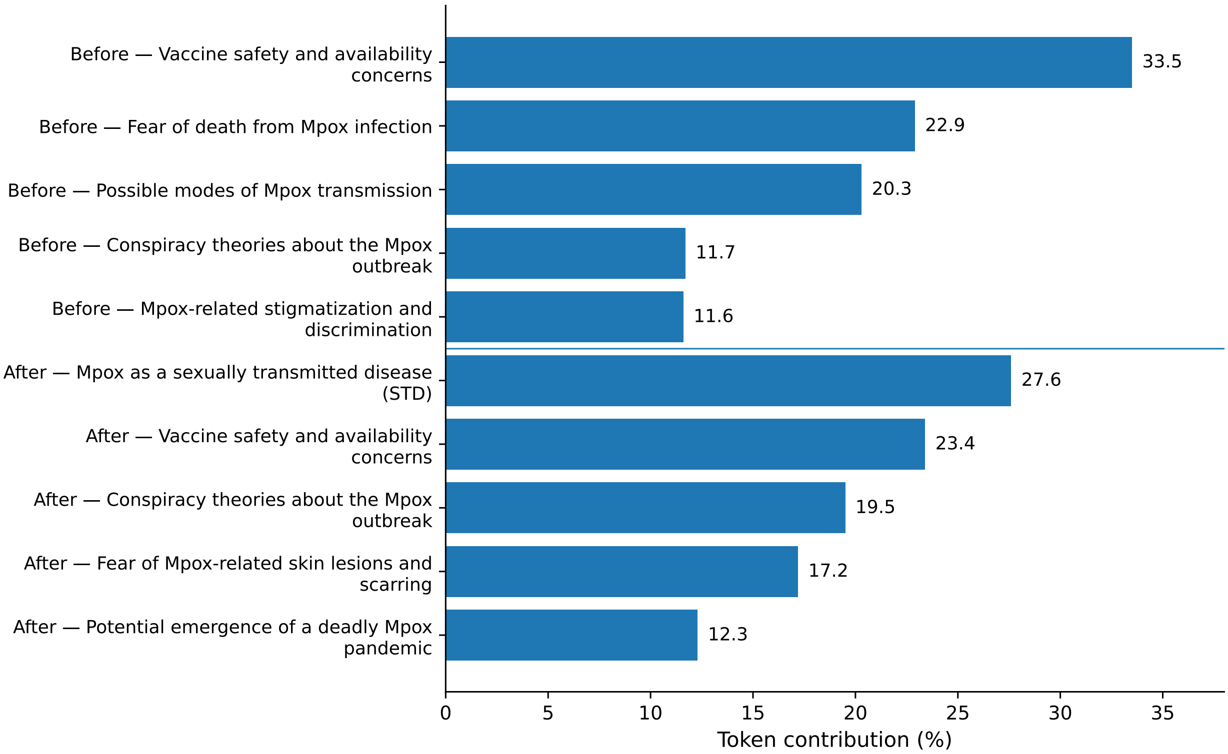

Figure 5: Dominant LDA-derived topics among Mpox tweets misclassified by COVID-19 RoBERTa before and after Mpox-specific fine-tuning, illustrating how domain adaptation altered the thematic structure of the remaining errors.

Model / stage	Precision	Recall	F1	Accuracy
COVID-19 BERT, 3-class	60	61	60	61
COVID-19 RoBERTa, 3-class	62	61	61	61
COVID-19 BERT, 2-class	81	79	80	79
COVID-19 RoBERTa, 2-class	82	79	80	79
Mpox VADER labels	49	50	47	50
Mpox TextBlob labels	41	40	37	40
NLP-Town BERT, before Mpox fine-tuning	43	39	40	39
Cardiff-NLP RoBERTa, before Mpox fine-tuning	40	40	39	40
COVID-19 BERT, before Mpox fine-tuning	49	47	45	47
COVID-19 RoBERTa, before Mpox fine-tuning	48	47	45	47
NLP-Town BERT, after Mpox fine-tuning	60	61	60	60
Cardiff-NLP RoBERTa, after Mpox fine-tuning	58	62	60	59
COVID-19 BERT, after Mpox fine-tuning	69	68	68	68
COVID-19 RoBERTa, after Mpox fine-tuning	70	69	69	69

Table 4: Overall precision, recall, F1-score and accuracy (%) for the principal COVID-19 and Mpox classification stages, highlighting the effects of neutral-class removal, cross-crisis transfer and Mpox-specific fine-tuning.

4 Discussion

4.1 Labelling Reliability

Taken together, the two studies indicate that reliable stance and sentiment classification in health-related social-media data depends strongly on the quality of the reference labels used for model development. The COVID-19 study showed why this is nontrivial: South African Twitter discourse contained slang, sarcasm, punctuation, emojis and other context-dependent expressions that could alter the intended meaning of otherwise similar text, motivating the use of iterative human labelling rather than relying solely on automatically assigned polarity. The Mpox study then tested this concern directly by comparing VADER and TextBlob against manually assigned stance labels and found that transformer models fine-tuned on the hand-labelled corpus ultimately outperformed both lexicon-based approaches. The combined evidence therefore supports manual labelling not merely as a preliminary data-processing step, but as a central source of domain-specific supervision for developing and evaluating models intended to interpret nuanced public-health discourse. [1, 2]

The iterative annotation process in the COVID-19 study further demonstrates that label reliability improved through explicit refinement rather than being assumed from the outset. Mean inter-annotator agreement increased from $\kappa = 0.67$ to $\kappa = 0.85$ after repeated review of disagreements and clarification of the labelling rules. This is especially important for borderline cases, including subtle hesitancy, where uncertainty, delayed willingness or ambiguous phrasing could reasonably be interpreted differently by annotators. The four qualitative error categories used across both studies–clear-cut, borderline, difficult-to-label and same-text cases–also showed that classification difficulty was closely tied to linguistic ambiguity rather than simply model weakness. In this sense, the reliability of the final labels depended on combining explicit decision rules with human contextual judgement, and the improved agreement provides empirical support for the resulting annotation framework. [1, 5]

4.2 Cross-Crisis Transferability

The comparison between the COVID-19 and Mpox studies provides evidence that model knowledge acquired in one public-health domain can remain useful when transferred to another, but that this transfer is incomplete without additional adaptation. When applied directly to Mpox, the COVID-19 BERT and RoBERTa models outperformed the generic pretrained transformer models, both reaching 45% F1 and 47% accuracy compared with 39–40% F1 and 39–40% accuracy for the independently pretrained alternatives. After fine-tuning on the manually labelled Mpox corpus, performance increased substantially to 68% for COVID-19 BERT and 69% for COVID-19 RoBERTa. These results suggest that prior exposure to related health discourse conferred a measurable advantage, while the additional gains obtained from Mpox-specific fine-tuning demonstrate that effective transfer still depended on new domain-relevant labelled data. [2]

The transfer results also suggest that what generalized across crises was not a complete stance-classification solution, but a useful representation of related health-discourse patterns that could be refined for the new domain. The COVID-19-adapted transformers began from a stronger position than the generic pretrained models, yet their initial Mpox performance remained well below that achieved after exposure to Mpox-specific labels. This distinction is important because it avoids interpreting cross-crisis transfer as a substitute for new annotation. Instead, the combined results support a layered view of adaptation in which prior domain-specific training provides an informative starting point, while local linguistic context, topic-specific concerns and new forms of ambiguity still require additional supervision. The Mpox error-topic analysis reinforces this interpretation by showing that fine-tuning altered the thematic structure of the remaining misclassifications rather than merely increasing a single aggregate score. [2]

4.3 Limitations

The principal limitations of both studies arise from the nature of social-media data and from differences between the two corpora. Twitter users do not constitute a representative sample of the broader population, with platform demographics potentially favouring younger, urban and more digitally engaged users, while keyword- and hashtag-based collection can introduce additional selection bias by determining which forms of discourse enter the dataset. These effects are particularly important when interpreting the COVID-19 results as indicators of vaccine hesitancy within South Africa and the Mpox results as indicators of wider outbreak-related attitudes. Moreover, the COVID-19 and Mpox corpora differed in geographic scope, collection period, search criteria and class distribution. They therefore provide a useful setting for examining methodological transfer under domain shift, but they should not be treated as matched samples or used to infer direct changes in public sentiment from one crisis to the other. [13, 14]

A second limitation concerns the construction and interpretation of the target labels themselves. Although manual labelling provided a stronger context-sensitive reference than fully automated polarity assignment, it did not eliminate subjectivity, particularly for neutral, subtly hesitant and linguistically ambiguous tweets. The COVID-19 study explicitly identified sentiment ambiguity–especially for neutral classifications–as a recurring challenge in social-media NLP, while the iterative increase in inter-annotator agreement reduced rather than removed the need for human judgement. The distinction between sentiment and stance introduces an additional constraint when comparing the studies: the COVID-19 work operationalised vaccine hesitancy largely through sentiment-oriented positive, neutral and negative labels, whereas the Mpox study more explicitly defined the task as stance towards a specific public-health target. The shared three-class framework therefore supports methodological continuity, but the labels should not be interpreted as perfectly identical constructs across the two applications. [1–3]

A further limitation is the dependence of both modelling pipelines on domain-specific data and task definitions. The COVID-19 study identifies potential bias from keyword selection, label definitions, data heterogeneity and neutral-class ambiguity, while the Mpox study notes that manually labelled datasets are labour-intensive and may themselves introduce bias. These constraints limit how confidently either model can be expected to generalise to different platforms, later stages of the same crisis, or unrelated public-health topics without further validation. The strong improvement observed after Mpox-specific fine-tuning therefore demonstrates the value of domain adaptation, but also highlights a practical scalability trade-off: high-quality human-labelled data can materially improve performance, yet producing such labels requires substantial time and expert judgement. [1, 2]

The methodological trajectory established across the COVID-19 and Mpox studies also informed later work applying NLP and machine learning in a substantially different scientific setting. Experience gained through manual labelling, text preprocessing, information extraction, model training, hyperparameter optimisation and performance evaluation provided a foundation for subsequent MSc research on ATLAS TileCal detector-control alarms, where semi-structured operational records were transformed into structured spatial and temporal datasets for predictive modelling. That later work, presented in the dissertation *Alert Signal Classification and Prediction Using Natural Language Processing for the Tile Calorimeter of ATLAS*, extended these computational skills from public-health social-media analysis to detector operations and predictive maintenance. The connection is therefore methodological rather than topical: the health studies established practical experience in extracting reliable information from complex text-like data and evaluating ML models, which was subsequently applied to a different class of scientific data. [17]

5 Conclusion

The combined evidence from the COVID-19 and Mpox studies demonstrates that effective stance and sentiment classification in emerging public-health discourse depends on both reliable human-labelled data and appropriate domain adaptation of machine-learning models. The COVID-19 study established a manually labelled framework for distinguishing overt hesitancy, subtle hesitancy and positive vaccine sentiment while showing that transformer models outperformed classical and recurrent alternatives, although neutral and linguistically ambiguous cases remained difficult to classify. The subsequent Mpox study extended this framework to a new health crisis, showing that COVID-19-adapted transformers transferred useful prior knowledge but achieved their strongest performance only after further fine-tuning on manually labelled Mpox data. Taken together, the studies therefore support neither purely automated labelling nor unmodified cross-domain model transfer as sufficient on their own; rather, they indicate that human contextual judgement and transferable machine-learning representations can operate as complementary components of a progressively adaptable NLP framework. [1, 2]

More broadly, the progression from COVID-19 vaccine hesitancy to Mpox illustrates both the value and the limits of methodological reuse across rapidly evolving health crises. Consistent labelling principles and previously adapted transformer models provided a useful starting point for the second study, but differences in language, topic, class distribution and public-health context required renewed human supervision and domain-specific fine-tuning. The resulting framework therefore favours adaptation rather than simple replication: transferable models can reduce the distance between related classification problems, while carefully constructed human labels remain essential for defining the target task, validating automated approaches and correcting for context-specific ambiguity. Future applications of health-related stance detection should consequently combine reusable NLP models with transparent labelling protocols and explicit validation whenever they are deployed in a new domain, population or stage of an emerging crisis.

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